

A Short History of the Grant Study

3

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The sign on the door read “The Grant Study of Adult Development.” Financed by W. T. Grant, the department store magnate, and run by Harvard’s Health Services Department, the study proposed to investigate “normal” young men, whatever that might mean.

On that particular afternoon, I was a sophomore, just turned nineteen. The Depression and a six-month siege of polio had been the sole departures from an otherwise contented, if not stimulating, life.

—BENJAMIN BRADLEE, *A Good Life*, 1995

In this chapter, I record the Grant Study’s seventy-five-year history for posterity—to be pondered, skimmed, or skipped at the reader’s pleasure. It’s the story not only of the Study, but also of seventy-five years of social sciences in America and the worldviews that came and went over that period. The research program of the Grant Study was directed by Clark Heath, M.D., from 1938 until 1954, by Charles McArthur, Ph.D., from 1954 to 1972, and from 1972 until 2004 by me. Since 2005, Robert Waldinger, M.D., has been the director of the Study.

In the academic year 1936–1937, Arlen V. Bock, M.D., became Oliver Professor of Hygiene at Harvard and chief of the student health services. In a report to President James Conant, he proposed broadening the scope of the Department of Hygiene and the role of the college physician. As a first step, he suggested a scientific study of healthy young men. Bock believed that health should be as much a central focus of medicine as pathology. As his proposal gained ground, he became more explicit about the kinds of issues he had in mind: the problem of nature vs. nurture; connections between personality and health; whether mental and physical illnesses can be predicted; how constitutional considerations might influence career choice. But his primary interest was: What is health? That question (and others that grew out of it) inspired the Grant Study for seventy-five years, and in this book I will start to answer it.

In his report to Conant, Bock cited “the stress of modern pressures” which, he felt, “the current generation of students had to face largely unprepared.”¹ And he thought that Harvard should be addressing these pressures.

As time passes, it seems logical to expect a different emphasis on the work of the Department of Hygiene than has been customary because of the growing complexity of human relations and the need to turn men out of the University better qualified to take their places in the affairs of life.²

To further this goal, Bock enlisted the support of his friend and patient William T. Grant, owner of the chain of stores bearing his name. Funding began on November 1, 1937, with the arrival of the first check from what would soon become the Grant Foundation, in the amount of \$60,000 (\$900,000 in 2009 dollars), and President Conant and the Harvard faculty approved Bock’s project.

Originally called the Harvard Longitudinal Study, it soon became known as the Harvard Grant Study of Social Adjustments. (This name reflected a major business preoccupation of Grant’s—namely, what makes a good store manager.) After Grant withdrew funding in 1947 the name was changed again, this time to the Harvard Study of Adult Development. But colloquially it has always been the Grant Study, and I will follow that convention here. (In 1967, when I was new to the Study, I naively asked why it was called this. A more senior investigator replied with a straight face, “Because it took an awful lot of grants to keep it going.”)

The Grant Study got under way in the fall of 1938, in a squat redbrick building on Holyoke Street in Cambridge, next to the Department of Hygiene. Its multidisciplinary aims were reflected in the composition of the original staff: an internist, a psychologist, a physical anthropologist, a psychiatrist, a physiologist, a caseworker, and two secretaries. Dan Fenn, Jr., editor of the *Harvard Crimson* at the time, said of the eight pioneers that they were “working on what might one day be one of Harvard’s important contributions to society, the analysis of the ‘normal’ person. . . . They may be able to

draw up a formula which will easily and correctly guide a man to his proper place in the world's society.” 3

In 1939, Harvard celebrated the opening of the Grant Study with a conference. Alas, I can find no record of what was discussed under its aegis, but it brought together a group of scientists of international distinction, who in their own ways shaped the Study profoundly. One was Adolf Meyer, who was the founding chairman of the department of psychiatry at Johns Hopkins and the new Study's patron saint. Meyer was perhaps the greatest advocate in America of a long-range view in psychiatry. He had come to this country in 1892 to look at the way the brain changes after death. Ten years later his interest had shifted from the neuropathology of the dead to the adaptive neurophysiology of the living. Meyer insisted that the study of psychiatry was the study of lives, and published a famous, if rarely read, paper on the value of the “life chart.” 4 In this he pleaded with his fellow psychiatrists for “a conscientious study of the mental life of patients,” insisting that “we need less discussion of generalities and more records of well observed cases—especially records of lifetimes—not merely snatches of picturesque symptoms or transcriptions of the meaning in traditional terms.” As the sagas of Adam Newman and Godfrey Camille have already made clear, the Grant Study made Meyer's dream a reality. Originally planned to last fifteen to twenty years—a mightily ambitious goal even today—it has now been making “records of lifetimes” for seventy-five years.

America's greatest physiologist, Walter Cannon, was at the conference too. Like Meyer, Cannon has been a role model for the Study throughout its existence. It was he who formulated the concept of the fight-or-flight response, and as a Harvard professor he wrote a classic monograph on physiological homeostasis, *The Wisdom of the Body*. 5 Psychological homeostasis has been an enduring concern of the Grant Study, and I chose the title of my book *The Wisdom of the Ego* in homage to Cannon. 6

President James Conant was present; he led Harvard through World War II, and served—importantly if less publicly—as civilian administrator of the Manhattan Project, where he opposed the development of the hydrogen bomb. Arlie Bock was there too.

Arlen Vernon Bock, once described by the *Harvard Gazette* as “blond, brisk, brusque, benign, belligerent, and always busy,” was a no-nonsense physician. 7 He had grown up one of eleven children on a farm in Iowa, and Harvard Medical School accepted him even though no one on the admissions committee had ever heard of his college. Bock began his career in the 1920s with a Moseley Traveling Fellowship for the study of medicine in Europe, following which he undertook a study of the physiological adaptation of men living in the high Andes. This experience led to his interest in physical fitness and positive health.

Bock and his colleague John W. Thompson (also at the conference) were pioneers in the study of normal human physiology, and in 1926 had helped found the Harvard Fatigue Laboratory, where physiologists, biologists, and chemists studied men's ability to adapt to physical stress. There they developed a technique of exercising subjects by having them step on and off locker-room benches, which is still part of modern cardiac testing. 8 The Laboratory was officially located on the campus of the Graduate School of Business Administration, but it sent teams traveling around the world, from the tropics of the Canal Zone to the Andean peaks. Its work eventually led to the decision of the United States Air Force to equip its new high-altitude bombers with supplementary oxygen.

Bock had an expansive vision, and he never stopped inveighing against medicine's tendency to think small and specialized. Writing to accept the directorship of the Harvard Department of Hygiene, he asserted that medical research paid too much attention to sick people, and that dividing the body up into symptoms and diseases could never shed light on the urgent question of how to live well. It was he who first conceptualized the concept of positive health; sixty years later, University of Pennsylvania psychologist Martin Seligman would take Bock's challenge into the new field of study we call positive psychology.

Even after the Grant Study got under way Bock's close affiliation with the Fatigue Lab continued, and it's worth noting that he himself walked two miles a day until his death at 96. He never forgot that “normal” and “average” are not the same thing—20/20 eyesight is normal, but unfortunately not average—and his interest was not in elucidating average fitness, but the best fitness possible. To accomplish this, it seemed sensible at the time to study an elite sample of men. And that's what the Grant Study did.

For its first seventeen years (from its beginnings in 1937–1938 until Charles McArthur assumed the directorship in 1955), the Study was dominated by its founder Arlie Bock, its first director, Clark Heath, M.D., and its social investigator, Lewise Gregory. It was the dedication and kindness of these three people that created the loyalty and gratitude that bonded the men to the Study until death did them part.

THE PIONEERS

Clark Heath, M.D., who served as director from 1938 to 1954, was a promising research scientist. He had once worked with Professor William Castle, a co-discoverer of vitamin B-12. He was also the staff internist, but his job description covered far more than that, and it expanded steadily over the years. He was responsible for the budget, for all necessary reports, for planning for the future, and for putting together case summaries of the Grant men. Still, it was not his administrative skills that made him so important to the young Study, but his clinical gifts.

The files of those early years make clear just how completely Heath epitomized the warm and caring physician. As each man joined the Study, Heath gave him an unusually complete two-hour physical exam. That was part of the Study routine. But until he left Harvard, returning Grant Study alumni came back to him voluntarily, seeking consultation for themselves and their families. When they needed more than mere medical advice, Heath saw a number of these ex-students for psychological counseling.

Lewise Gregory (later Davies), a perceptive Virginian, was the third member of this key trio. Bock needed a social investigator to interview all the Study members and their families. Gregory's only professional education was secretarial school, but Bock chose her for her dazzling interpersonal skills and her innate talent as an interviewer. After her death, Charles McArthur, the second director, described her technique for a staff member who was writing a brief (unpublished) history of the Study. "When she would visit a student's family, she would sit daintily with her legs crossed, affix her large blue eyes on the conversationalist, and the smitten parents would bare their lives to her." She was an attentive and sympathetic listener, and the loyalty she engendered among the participants and their families would last a lifetime. She also endeared herself to the Study men as a big sister. (A very pretty one. And her sister was Margaret Sullavan, the movie star.) Miss Gregory was invaluable in bringing lost sheep back into the fold, and the Grant Study's attrition rate—the lowest of any similar study in history—is a tribute to her diligence and her diplomacy.

In the beginning, Gregory's job was to take a careful social history from each subject as he entered the Study, and then interview his parents. In those early years she traveled the length and breadth of the country to meet the men's families in their homes. She obtained from the mother a detailed history of the early development of each Study man. She also took a family history that included descriptions of grandparents, aunts, uncles, and first cousins, and of any mental illnesses in the extended family. I encountered some of these family members many years later, and they still recalled her visit with great pleasure.

In her family interviews, Gregory accentuated the positive. The usual psychiatric social history with its emphasis upon pathology tends to make all of us look like fugitives from a Tennessee Williams play. Gregory asked about problems in the men's growing up, but she also wanted to hear about what had gone well. Not only did this foster the alliance between the Study and its members, but it also meant that when pathological details did come up in the social histories, they were usually significant.

After her great work of 268 family interviews was complete, Gregory remained with the Study part time. Even in the 1980s, at the end of her tenure, she was still the one who could charm nonresponders back into continued participation—which she did with astonishing success.

William T. Grant, of course, was an important figure in the founding of the Grant Study. Billy Grant, as his friends called him, was a tenth-grade dropout who founded a store for low-priced household wares in 1906. Nothing in his first store cost more than twenty-five cents—he prided himself on that—but it grew into the Walmart of the 1930s. Grant's foundation has become world-famous, too, but it got its start in that first gift to Arlie Bock in 1937. The two men had different visions. Bock hoped that his research into an optimally healthy population would help the United States military select better officer candidates; Grant hoped that his new foundation's first project would help him identify effective managers for his multitude of chain stores. They both hoped to identify superachievers. But Grant was more interested in social and emotional intelligence than the other Grant Study investigators, who shared the contemporary preoccupation

with constitutional medicine. By 1945 this would lead to friction. But until the end of the war Bock maintained a close relationship with Grant, visiting him at his houses in Florida and Connecticut.

Frederic Lyman Wells, Ph.D., the Study's chief psychologist, was a psychometrician and one of the developers of the Army Alpha Tests, the major test for intelligence screening in World War I. Wells came from a distinguished New England academic family. He started college at fifteen, and by twenty had earned his M.A. From 1925 to 1928 he served on government advisory boards, such as the National Research Council and the National Committee for Mental Hygiene. His Grant Study brief was to determine the personality organizations, interests and aptitudes, and intelligence levels of the Grant Study subjects. Wells was probably the most distinguished scientist on the staff, and he gained further eminence as a consultant to the War Department between 1941 and 1946, when he helped to develop the Army General Classification Test (a measure of intelligence and vocational skills). He was a hard worker and a methodical and systematic analyst, but given to dry and lengthy statistical expositions. His reports reveal little of the rich humanity of the Study men.

Carl Seltzer, Ph.D., a young physical anthropologist who had worked closely with Earnest Hooton and William Sheldon (on whom more below), was another exponent of constitutional medicine, specifically the relationship between body type and personality. John W. Thompson, Ph.D., a Scot and co-founder of the Fatigue Lab with Bock, and then Lucien Brouha, Ph.D., a Belgian refugee from war-torn Europe, served as early physiologists to the Grant Study staff. Both were gone by 1943, however, and funding for the Fatigue Lab seems to have halted by 1944. Vicissitudes in funding are one of the ways we know when scientific fashions begin to change. Constitutional medicine was on the way out.

The Study's staff psychiatrists were responsible for extensive interviews of the Study members—about ten hours each. None of the five men who filled the position in the Study's early days stayed longer than three years. Two of them went on to distinguished academic careers: Donald Hastings (1938) as chairman of the Department of Psychiatry at the University of Minnesota Medical School, and Douglas Bond (1942) as dean of Case Western Reserve Medical School. Psychiatrists William Woods (1942–45), John Flumerfelt (1940–41), and Thomas Wright (1939–40) also served as interviewers. Woods was also responsible for an assessment scheme of twenty-six personality traits that underpinned much of the early research. 10

Unfortunately, however, the early staffing and implementation of the Grant Study did not reflect the early work of four important investigators of personality. Their work profoundly affected my own later interpretation of the Study's data, but in 1937–42 it was still too new to inform the Study itself. Heinz Hartmann, Anna Freud, Erik Erikson, and Harry Stack Sullivan all significantly shaped the modern understanding of healthy personality. The first three were instrumental in replacing Sigmund Freud's view of personality as an often-pathological compromise between cognitive morality (superego) and irrational passion (id). Hartmann, Anna Freud, and her student Erikson offered an alternative conception of personality as a product of involuntary, but usually healthy and creative, adaptation. Anna Freud's *The Ego and the Mechanisms of Defense* was first published in 1937; two years later, Hartmann published his own classic work on ego psychology, which would appear in English as *Ego Psychology and the Problem of Adaptation*. However, it wasn't until 1967 that the Grant Study began to focus on the men's styles of psychological coping (see Chapter 8).

Harry Stack Sullivan was another pioneer, who opened psychiatry up to the science of relationships that John Bowlby and his student Mary Ainsworth would make famous—but not until the second half of the twentieth century.

FROM BIOLOGY TO PSYCHOLOGY

Let me expand on these omissions for a bit, to show how time changes both science and scientists. It is easy to forget how very recent is our current interest in intimate relationships. For the first ten years of the Study, biological theorizing held undisputed pride of place. In 1938, constitution and eugenics (as in breeding) were considered far more potent forces than environment in how people turned out. Biological indicators were tracked in minute detail, but it was the rare social scientist who paid any attention to what would become known as “emotional intelligence,” particularly the capacity for love and close friendship. Harry Harlow was one of these, a psychologist/ethologist who earned renown for groundbreaking studies of relationship deprivation in monkeys. In his 1958 presidential address to the American Psychological Association, he felt driven to lament, “Psychologists not only show no interest in the origin and development of love and affection, but they seem to be unaware of its very existence!” 11

At the time of Harlow's speech, behaviorists like B. F. Skinner and John Watson were assuming that babies became attached to their mothers because their mothers fed them. Psychoanalysts Sigmund and Anna Freud believed essentially the same thing. However different the behaviorist and psychoanalytic psychologies, they shared a rather concrete view of the interplay between biology and emotion. Lust, hunger, and the vicissitudes of power ruled the psychological universe. Love was conceptualized as Eros—a matter of individual hedonistic instinct, not a process of reciprocal pair bonding,

It was only in 1950 that John Bowlby, who was both a psychoanalyst and an ethologist, began to establish awareness that attachment experiences are fundamental shapers of personality, and that babies "imprint" on their mothers not because their mothers fill their bellies, but because they cuddle them, sing to them, and gaze into their eyes. Experimental evidence soon followed. But I can attest that years after the Grant Study began, English teachers were still drilling the youth of the 1940s on Rudyard Kipling's Victorian mantra "He travels the fastest who travels alone." A relational world governed by oxytocin, mirror neurons, and limbic maternal attachment (a.k.a. love) was inconceivable in the psychology of the Study's first decade.

There is an interesting parallel here with infantile autism. This fairly common disorder, which is due to a congenital absence of empathy, was not spotted until 1943, when a child psychiatrist finally noticed it in his own son. Its close relative, Asperger's syndrome, was identified in 1944. But it was fifty years more before those genetic disorders were absorbed into psychiatry's diagnostic framework. In other words, in the 1930s, the congenital impairment of attachment reflected in childhood autism was harder for scientists to grasp than quantum physics. The functional reality of relatedness had not been incorporated into the consciousness of the social sciences.

Cultural anthropology, pioneered by Franz Boas, captured the hearts and minds of college students in the sixties, but it was a relatively esoteric discipline in 1940, and physical anthropology still dominated. Ernst Kretschmer, a German psychiatrist, was nominated in 1929 for the Nobel Prize in medicine for his work on body build and character. 12 Investigators inspired by him believed that personality was determined by somatotype—three physical bodily conformations (thin, frail ectomorphs, robust muscular mesomorphs, and soft, plump endomorphs). Social scientists still believed that the British Empire was the result of genetic racial superiority, not the environmental luck of owning the "guns, germs and steel" that Jared Diamond made famous in the brilliant book of that name, in which he proved once and for all that racial dominance was a product of culture and geography, and not of biological heredity.

So while the biological determinism of the early Grant investigators sometimes sounds like racism to twenty-first-century ears, it had less to do with cryptofascism than with the absence of an alternative paradigm. Many hours were spent assessing the physiology, somato-type, and "racial typology" of the Grant Study sophomores. During their ten hours of psychiatric interviews they were asked about masturbation and their opinions on premarital sex. But nobody asked them about their friends or their girlfriends. Until the 1970s at least, attachment and empathy were the province of sentimental novels, not of science. A pity.

THE MEN OF THE GRANT STUDY

As of this writing, all but seven of the surviving members of the Grant Study have reached their ninetieth birthdays. When the Study began, however, they were college sophomores, most of them nineteen years old. The original College cohort—the group of Grant Study men, who liked to refer to themselves as "the guinea pigs"—numbered 268. Sixty-four men were drawn from the Harvard classes of 1939, 1940, and 1941, and 204 came from a more systematic 7–8 percent sample drawn from the sophomore classes of 1942–1944.

About 10 percent of the men ended up in the Study by chance: a few volunteers; some younger brothers of already participating men; the occasional student referred by an advisor. The other nine-tenths of the sample were selected according to the following rough protocol, of which the details varied slightly from year to year.

First the Study investigators went through the new class, screening out any students who might not graduate. On the advice of the dean of students, they used as criteria the students' SAT scores in combination with grade point average and observed measures of natural ability. A high school valedictorian with modest SATs would be preferred to a boy who tested well but whose class standing in high school was low. These criteria eliminated 40 percent of each Harvard entering class.

Known medical or psychological difficulties led to the exclusion of another 30 percent. The (roughly 300) names remaining were submitted to the college deans, who picked from among them about 100 students each year whom they perceived as “sound.” In essence, this meant students the deans were glad they had admitted, especially the ones involved in extracurricular activities and freshman athletics. For a single class year, the Grant Study selected the men who would become editor-in-chief of the *Crimson*, president of the *Advocate* (the college literary magazine), and president of the *Harvard Lampoon*. Four times as many of the Grant Study men as chance would have predicted held class offices, both in college and, as it turned out over the next half-century, at college reunions. However, the deans also chose a disproportionate number of “national scholars,” gifted men from poor families for whom all expenses, including transportation, were provided by Harvard. These youths, often relatively inept socially, were chosen solely for potential academic brilliance.

The freshman physicals of the classes as a whole tell us—and it’s no surprise—that the Grant men were twice as likely as their classmates to be muscular mesomorphs, rather than skinny ectomorphs or pudgy endomorphs.

Out of the 90 (10 percent of each class) sophomores chosen, roughly 1 in 5 ended up not joining the Study for reasons of his own (schedule conflict, disinclination, or failure to show up for intake procedures). About 70 men a year were added to the Study from the classes of 1942 through 1944, bringing the total College cohort to 268.

The Grant Study deliberately cast its net for men who were likely to lead “successful” lives. All of them had already been selected for admission to a competitive and demanding college, and then they were selected further for their capacity to master college life and, in the words of Arlie Bock, “paddle their own canoe.” Many of them were firstborn sons, and independent men were preferred over less autonomous ones. The Study selectors were looking for men with the capacity to live up to or exceed an already high level of natural ability.

The men of the Grant study were homogeneous in many ways. They were well matched in physical and mental health, skin color, education, intellect and academic achievement, and culture and historical epoch (see Table 3.1). They all lived through the Great Depression, and they all shared the likelihood of active participation in World War II in the immediate future.

The Study administered the Army Alpha Intelligence Tests, which put the IQs of most of the men in the top 3 percent of the general population. Their College Board SAT scores (average 584) put them among the top 5 to 10 percent of college-bound high school graduates, but not beyond the range of many other able college students. Of course, in 1940 the population taking the SAT was more select than today; still, many readers of this book will be entitled to sneer, “Well, I scored far better than that.”

Table 3.1 The Historical Setting of the Grant Study Birth Cohort

The average Grant Study man was fifth-generation American. A few were immigrants, but more than a few had families that had been here for ten generations. There were no African Americans. Ten percent of the men were Catholics and 10 percent Jews. The remaining 80 percent, a higher percentage than among their classmates, were Protestants. Eighty-nine percent of the men came from north of the Mason-Dixon Line and east of the Missouri. Twenty-five years later, 75 percent of the sample remained within these boundaries, and 60 percent of the total sample had migrated to the five urban meccas of San Francisco, New York, Washington, Boston, and Chicago.

They were mostly a privileged group, but here they were less perfectly matched. Half of the men had had some private education, but often on scholarship. In college, 40 percent received financial aid (at that time, a year at Harvard cost about \$22,500 in 2009 dollars), and half worked during the academic year, paying college costs not covered by their scholarships themselves. The men’s families were classified on the basis of paternal education, occupation, and selection for *Who’s Who* in America. One-third of the men’s fathers had had some professional training, but half of the parents had no college degree. Only 11 percent of the men’s mothers had ever worked, and of those who had, most had been single parents. Of the thirty-two working mothers, two were writers, five schoolteachers, one an artist, and one a lawyer; the rest were secretaries and waitresses.

Once Lewise Gregory began her home interviews, the families were also classified more subjectively by reference to such class- and status-related markers as household furnishings, books, art, and size of house. There was significant variation there. Sixteen percent of the families were categorized as upper class. Even during the Depression they enjoyed

multiple houses, motorcars, and servants. Their mean yearly income was \$225,000 per capita—yes, I mean per family member—in 2009 dollars. Four percent of the families were classified as lower class. Their annual mean per capita income was \$5,200 in 2009 dollars.

So the men did not all come to college with silver spoons in their mouths; and even when they did, their parents or grandparents might have had more humble beginnings. There was Alfred Paine, for example, whom we will get to know more extensively in Chapter 7. He had a trust fund from birth; his father had been a head of the New York Stock Exchange; his grandfather was a successful merchant banker. But that grandfather had made his first thousand dollars as an itinerant pioneer, picking up buffalo horns on the Great Plains at night and shipping them back to New England for resale.

The father of another study member, Brian Farmer, was a painter and paperhanger. Soon after Brian's birth, work grew so scarce that Mr. Farmer moved his family to South Dakota, where he and his wife and older children worked in the sugar beet fields as laborers. For their combined efforts, the family received eleven dollars for every acre they cleared. They had barely enough to eat until a kindly neighbor told them that they could have all the beans and potatoes they wanted from his fields. Together they gathered enough beans and potatoes to last them through the winter, plus a surplus that they traded for staples such as sugar, salt, and other groceries. During those years the Farmers did not know what it was to taste fresh vegetables or fruits. Mr. Farmer picked up odd jobs here and there, but his neighbors were too poor to pay him in cash. When Brian entered Harvard, his father was still earning only five dollars a day.

Some of the men had other kinds of difficulties. One Study member, whom his adult friends knew as “a very happy guy,” talked about growing up with an alcoholic mother. “I had a terrible time in my second, fourth and sixth grades. I got trial promotions probably because the teacher wanted to get rid of me. Throughout this period, also, there was hardly any income in the family. My folks had lost the variety store and gas station and during the winter, warmth at night consisted of getting under a pile of blankets. The winter days of my early years were spent curled up on a bench behind a big potbelly stove in a variety store listening to men talk. I would stay there because it was warmer than it was in the house.”

Whatever a man's origins, his Harvard degree was a ticket of entry to the upper middle class. When the men returned from their service in World War II, they also benefited from high employment, a strong dollar, and the G.I. Bill, which virtually guaranteed an affordable graduate school education. They were just young enough to participate in the physical fitness and anti-smoking trends of the sixties, seventies, and eighties. Most of the 1940 generation of Harvard men were upwardly mobile, and most ended up more successful than their fathers. (There were a few exceptions, like the man whose Wall Street father made two million 1935 dollars a year in the midst of the Depression. No, this father was not Joseph P. Kennedy.)

HOW THE MEN WERE STUDIED: 1939–1946

Three investigators interviewed each sophomore accepted into the Study: Clark Heath, a staff psychiatrist, and Lewis Gregory. The meeting with Dr. Heath included a comprehensive two-hour physical and a history of the young man's dietary habits, medical history, and physical responses to stress. Each man was also studied by Carl Seltzer, the physical anthropologist, who recorded his racial type (Nordic, Mediterranean, etc.) and his somatotype (mesomorph, endomorph, ectomorph), determined whether his body build was predominately “masculine” or “feminine,” and made exhaustive anthropometric measurements. 13 The Study investigators noted his every physical detail, from the functioning of his major organs to his brow ridges and moles to the hanging length of his scrotum. They took a careful dietary history, too, including the number of teaspoons of sugar in his daily coffee or tea (the range was 0 to 7!).

As I've noted, the emphasis on somatotyping was an ill-fated effort to advance the then-fashionable science of physical anthropology. Carl Seltzer's mentor was the Harvard anthropologist William H. Sheldon, who had been influenced by Kretschmer's work and believed that human personality was significantly correlated with body build. The ectomorphic build, that is, was thought to be correlated with a schizoid temperament, the mesomorphic build with a sanguine temperament, and the endomorphic build with a manic-depressive temperament. 14 This is a cat that I let out of the bag in the last chapter; a third of a century later, Grant Study follow-up revealed that none of the ten outcomes in the Decathlon of Flourishing (or the men's officer potential) correlated significantly with body type.

Eight to ten one-hour psychiatric interviews focused on the man's family, his values, his religious experience, and his career plans. The psychiatrists were trying to get to know the men as people, not as patients. No attempt was made to look for pathology or to interpret the men's lives psychodynamically. The interviews included a history of early sexual development, but unfortunately the psychiatrists did not inquire into the men's close relationships.

Lewis Gregory integrated individual interviews with the men with the careful social histories that she gathered in her home visits with the men's parents; I've described these above. In keeping with the research methodology of the 1930s, those histories were sometimes more anecdotal than systematic. Her chief informants were usually the men's mothers, although fathers and siblings at times contributed information as well.

Frederic Wells, the Study psychologist, gave each man tests designed to reflect native intelligence (the Army Alpha Verbal and Alpha Numerical). In many cases he also administered two projective tests: a word association test and a shortened version of the Rorschach. The intention here was to test imagination, not to explore the unconscious (for which purpose the Rorschach is usually employed). Also included was the Harvard Block Assembly Test, an assessment of manipulative dexterity and comprehension of spatial relationships. 15

Physiologist Lucien Brouha studied each man in the Fatigue Lab. Brouha measured each man's respiratory functions and the physiologic effects of running on an 8.6 percent incline treadmill at seven miles per hour for five minutes or until exhausted, whichever came first. Examiners took measures of pulse rate, blood lactate levels, exercise tolerance, and so on as a way of sorting the students by physical fitness. Surprisingly, I noted in 2000—more than fifty years later—that treadmill endurance correlated better with successful relationships than with physical health. (As it turned out, endurance and stoicism turned out to be better predictors of love than health in other areas, too.)

In 1940, the Study received a one-time grant of \$2,400 (\$35,000 in 2009 dollars) from the Macy Foundation, a philanthropic organization sympathetic to psychosomatic medicine. This windfall enabled the recording of primitive single-channel electroencephalograms, which had just begun to come into use. The neophyte encephalographer's interpretations of these EEGs sometimes sound more like Tarot card readings than physiological analysis; in some cases the tracings were thought to reveal "latent homosexuality." The Study also hired an experienced forensic graphologist to interpret the men's handwriting. It soon became clear that neither handwriting nor EEGs were useful predictors of personality. But however naive or even comical some of these efforts sound now, they reflect a serious and important truth—that the Grant Study was collecting information even before there was science available to capitalize on its investment, in hopes that it would prove meaningful later. And in many cases it did. Prewar psychological and medical science were very different from today's counterparts. It wasn't only attachment theory that had to wait until the nineteen-fifties; so did double-blind placebo-controlled drug trials. Good science is always reaching a little ahead of itself.

Table 3.2 The Sequence of Contacts

1938–1945

8–10 psychiatric interviews

Complete physical exam by Dr. Heath

Interviews by Dr. Heath and Miss Gregory; she also made home visits

Anthropological and physiological testing

EEG, Rorschach, and handwriting analysis (on many of the men)

Complete psychometric testing and some projective testing by Dr. Wells

The 26 personality traits assigned by Dr. Woods (Appendix E)

Brief childhood assessment (1–3) by Miss Gregory

At age 21 the ABC College adjustment (soundness) ratings assigned

1946–1950

Debriefing of combat experience by Dr. Monks

Annual questionnaires begun

Interviews with wives by social anthropologist Dr. Lantis

Thematic Apperception Test by Dr. McArthur

Full staff conferences on each man at age 29; ABCDE ratings of personality soundness for future adjustment made 1950–1967

Mostly biennial questionnaires; little other contact

1967–1985

All men re-interviewed, mostly by Eva Milofsky or Dr. Vaillant (Appendix A) Beginning age 45, complete physical exams every five years until the present

Objective health scored 1–5 (Well, minor illness, chronic illness, disabling illness, dead)

Childhood environment assessed, blind to events after age 19

Wives twice sent questionnaires

Adjustment at work, love, and play assessed from age 30–47 on all men (Appendix D)

Adaptive coping style (narcissistic, neurotic, empathic) assessed

NEO (Costa and McCrae) administered by mail

Lazare Personality Inventory administered by mail

1985–2002

All men re-interviewed (Appendix A)

Adjustment at work, love, and play assessed from age 49–65 and from age 65–80 (Appendix D)

Wives and children sent questionnaires

Physical exams every five years and biennial questionnaires continued

Gallup Organization's Wellsprings of a Positive Life administered by mail

Aging at 80 computed (subjective and objective mental and physical health)

2002–2010

Physical exams every five years and biennial questionnaires continued

Joint marital interviews and "daily diaries"

Telephone Interview for Cognitive Status (TICS) at ages 80, 85, and 90

Re-interviewed at retirement, 1985–2005

2010

Decathlon scores compiled

Case in point: the scientific pendulum has swung once more, and in the early twenty-first century genetic research once more dominates studies of environment. Instead of handwriting samples, the Study is collecting DNA. We don't yet know how it will be used. What will that look like to readers seventy-five years from now?

EARLY DATA ANALYSIS

By 1941, the Study had examined 211 sophomores. A debate arose as to whether this was enough. Frederic Wells, perhaps the most established researcher employed by the Study, sent a letter to Clark Heath imploring that he stop accepting new students so that the staff could begin evaluating the huge amount of data that they already accumulated. But others worried that 211 cases was only a small number, after all, when the goal was to find conclusive correlations between the psychological and anthropological. Eventually a compromise was reached, and the researchers agreed to take on one last group of students from the class of 1944. They were studied as sophomores in 1942 and completed the total Study cohort of 268. Between 1938 and 1943 the Grant Foundation had contributed \$450,000 (\$7 million in 2009 dollars) to the research.

But now the Study had to decide how to harvest this vast planting, and that was not proving easy. As I've said, a longitudinal study has to stay a bit ahead of itself, both in its data collecting and in its theoretical hunches. After all, it can't know in advance what will prove relevant (if it did, there would be no need for the study), and has to rely on best guesses. This can be an advantage as well as a disadvantage. But the Grant Study was a very early longitudinal study, so it didn't have the benefit of others' experience, and it began in something of a theoretical vacuum, when very little was known about adult growth and development. It had been very ambitious in collecting data, but not always very thoughtful about what it would do with the results. Many of the investigators were feeling overwhelmed, and by 1944 William Grant was beginning to express serious doubts about the administration of the Study and about whether he should continue to fund it. The trustees of the Grant Foundation put increasing pressure on the researchers to produce a summary of the Study's early results. For the next two years, the Study archives are filled with desperate efforts to discover publishable material.

Only three papers appeared before 1945 (see Appendix F), and in 1945, two monographs were hastily published to meet Grant's demand for results and a popular book that would dramatize the Study's findings.

Earnest Hooton was a Harvard professor, a brilliant physical anthropologist, and a fluent writer. He had been Study anthropologist Carl Seltzer's mentor (also my archaeologist father's). And he was firmly committed to the world of constitutional medicine. "If we wish to study the whole man, we must begin with his physique," he maintained, and he imagined that research out of the Grant Study might one day lead to "effective control of individual quality through genetics, or breeding." 16 It was he who had encouraged early researchers to expect that "[O]n the whole . . . 'normality' goes closely with a 'strong masculine component.' " 17 Hooton's summary of the Grant Study was published in 1945 as *Young Man, You Are Normal*. 18 That was a far cry from Grant's own proposed title, *The Grant Study of Social Adjustment*, which did not help to bridge the growing rift between the Study administration and its funder. Still, for the next thirty years, Hooton's was the leading book on the study.

In it he wrote, "When physique, studied from different standpoints, turns out to be so intimately related to various personality traits, it is clear that body build must also furnish clues to the social capacities of the individual." 19 Instead of adducing any experimental evidence in support of these claims, however, he simply dismissed his opposition as "crass environmentalists." 20 This is a striking term. It conveys not just simple cluelessness about the environmental considerations that are so crucial now, but real antagonism to an entire way of thinking. Yet Arlie Bock had already been wondering about the nature/nurture issue in his first visions of the Study seven years before; it's possible to see in the difference between his attitude and Hooton's a scientific paradigm nearing its tipping point.

The second monograph was *What People Are*, by Study director Clark Heath. Heath's book relied heavily on anthropometric measurements too, and also on an untested personality profiling scheme devised by William Woods, a staff psychiatrist without research training. 21

Woods's schema (see Appendix E) scored Study members on twenty-six personality traits, many of them dichotomies such as Vital (warm, expressive) Affect vs. Bland (colorless) Affect, Well-Integrated Personality vs. Unintegrated Personality, Verbal vs. Inarticulate, Sociable vs. Asocial etc. An attempt was then made to correlate the resulting profiles with body build but the results were unconvincing.

Neither of these two hastily published books attracted much attention, and certainly they did nothing to halt the shift toward environmentalism in the social sciences. In the years since, somatotype categories have not proven particularly useful when matched against other independent ratings, and neither have (most of) Woods's personality traits. Even at the time, what evidence there was, was not persuasive. Worse, the early investigators were not blind to each other's ratings,

so there's no way to know how much apparent early correlations between body build and personality were a function of halo effect or observer bias. In 1970 I tried to replicate the correlations reported in some of the early papers and could not.

Before we contemptuously (and prematurely) dismiss the early investigators' work as a wash, however, we have to consider the realities of statistical methodology in those days. There were no computers to absorb infinitudes of entries and magically align them along dozens of axes at the click of a key. Study members had to be listed down the left-hand side of huge ledger sheets. Scores and measurements were plotted along the top, individually, in exquisite handwriting. And that was only the beginning. To be useful, the data had to be pulled out of the ledgers, again by hand, and subjected to individual calculations, some of them quite complex. The tests that the original investigators used are out of fashion today, but analysis was laborious in a way that is unimaginable now. The Study in the 1940s did not even own a Monroe calculator (as the early electric adding machines were called). Ironically, in 1944, one of the world's first reliable computers, the Mark I, was being installed only 300 yards away. But the techniques and technologies of complex data analysis that could (and in time did) reveal unexpected correlations in the early data were not yet available to the Grant Study researchers. When the rich crop they had sown finally did come to fruition after the explosion of computer technology in the late twentieth century, it was the early investigators who had done much of the heavy lifting.

EARLY CONCLUSIONS AND SOME CORRECTIONS

The specter of World War II engaged early the Study's interest in possible military applications for its research, particularly the identification of potential officers and appropriate placements for them. John Monks was a patrician internist who joined the staff in 1946 to study the men's response to the war, and produced a well-researched if little-appreciated monograph, *College Men at War*. 22 Monks treated Woods's personality traits as if their scientific validity had been proven, with the suggestion that they be used to identify desirable traits in officer candidates. His monograph was fascinating in the stories and backgrounds it gave of the College men. It was disappointing, however, because Monks, like the original investigators, had not yet grasped the power of longitudinal study to actually test a hypothesis.

And the fresh empirical follow-up of 2010 (which came out of the questions I was trying to answer for this book as well as the Decathlon challenge) found that such promising variables as Masculine Body Build, Well Integrated, Vital Affect, or such foreboding ones as Lack of Purpose and Values and Shy, bore no relation at all to attained military rank. The only college trait that did predict high rank was Political, while low rank was predicted only by . . . Cultural and Creative and Intuitive! The scriptwriters of M*A*S*H might have anticipated that result, but Bock's original investigators certainly did not.

Woods's predictive schema has had more general problems than its failure to identify good officer material. Most of his traits failed to correlate significantly with any of the events in the 2010 Decathlon of Flourishing, and only one correlated significantly with more than three. That one, however, was a notable exception. Well Integrated (defined as "steady, dependable, thorough, sincere, and trustworthy") correlated with eight Decathlon events, and for the last twenty-five years has been a staple variable in our data analyses. It signals a bundle of traits that enable a young man "to surmount common problems which confront him such as career choice, competitive environment, and moral and religious attitudes." 23 It was assigned to 60 percent of the men, while 15 percent were called Incompletely Integrated. These latter men were deemed to lack perseverance, and were seen as "erratic, unreliable, sporadic, undependable, ill directed and little organized." (The remaining 25 percent of the men were unclassified for this dichotomized variable.) Half a century later, proportionately four times as many of the Well Integrated enjoyed good marriages than the Incompletely Integrated. Another finding of great interest was that as of 2012, the Well Integrated have lived on average seven years longer. In contrast, the two variables that the original investigators thought would be most predictive of positive outcome, Vital Affect and Sociable, were unimportant after the first ten years.

TROUBLE IN PARADISE

Writing to Bock in 1944, Grant chastised him for managing "a cooperative study with a good deal more distrust, disdain and independence of outsiders than I have ever seen productive of satisfactory results." 24 He would be willing to contribute to the Study in the future, he said, only if Bock also secured additional funds from the Harvard Department of Hygiene. He promised one more payment of \$30,000 (\$300,000 in 2009 dollars) to fund the project until he and the trustees of his foundation could feel confident that the Study was organized effectively. In hindsight, it appears that no

one in the Study had fully understood the constraints of longitudinal studies, or the patience that they require. And while Clark Heath was a model of clinical expertise and personal warmth, administrative organization was not his forte. Ironically, there were gifted future administrators at the Grant Study, and one, at least, had an intuitive understanding of the powerful but demanding nature of this kind of research. Donald Hastings, once a Grant Study psychiatric interviewer, carried out and harvested one of the first well-designed long-term studies of neurotic outpatients when he was chairman of psychiatry at the University of Minnesota Medical School in 1958. 25 Hastings's study was one of the first prospective longitudinal studies ever published in the *American Journal of Psychiatry*, and the first that I ever clipped out of a journal as a medical student to keep in my own files. It was Hastings who showed me how prospective longitudinal study could bring order to muddled psychiatric thought, and I admired his work for years before I ever heard of the Grant Study.

Bock grumbled to Earl Bond, Grant's closest scientific advisor and future Grant Foundation trustee, "I had to tell Mr. Grant that the trustees of the foundation could not expect to run the Grant Study." After this, not surprisingly, Grant Foundation largesse dried up for almost twenty years. The embryonic Grant Foundation had no medical researchers or clinicians among its trustees, who did not yet understand either the limitations or the potential of the Grant Study's longitudinal design. An orchard like the Grant Study takes time to bear its fruit.

Even as publications began to emerge more regularly (Appendix F), the Study authors clung to the bounds of their respective disciplines, selecting the kinds of narrow topics that yield academic journal papers. They were not yet exploiting the prospective, longitudinal potential of the Study. For example, they could have tested the somato- type data more critically; certainly they had the necessary material to examine in 1946 whether this information did or didn't predict anything about military rank at discharge. But they didn't. By 1970, even after fifty publications, the Grant Study remained little known and little cited. It had also been underfunded for fifteen years, as I will describe shortly.

On the other hand, the seeds that Grant, Bock, and Heath had planted really did promise a rich harvest decades into the future. For example, as we'll see in Chapter 10 , Woods's intuitively derived personality traits permitted us to predict political choice reliably over the next half-century of the men's voting lives. And forty years later, Monks's careful debriefing of the men after the war regarding the details of their combat experiences led to one of the world's few prospective studies of the antecedents of posttraumatic stress disorder. 26

THE WAR AND ITS AFTERMATH

After Pearl Harbor, most of the Grant Study men went into the armed services directly from Harvard, and most of them performed unusually well there. But they took their first real steps into adult life in a world reeling from the stupendous social and economic sequelae of the war, which made rigorous demands on their adaptive capacities.

THE NEXT TWENTY-FIVE YEARS: 1945–1970

The end of the war saw a sea change in Western social science as well as in Western society. Hooton's ferocity notwithstanding, genetics and biological determinism were out, partly out of emotional overreaction to the Nazi misuse of eugenics in pursuit of a racist ideology. Relativism, Skinner, and cultural anthropology were in, and scientists were looking beyond constitutional endowment for the conditions of life that shape individual outcomes. In the United States, psychoanalysts were becoming chairmen of major departments of psychiatry.

Censorship issues during the war years limited communication between the Grant Study and the men to the occasional V-mail. But once the war was over, the Study began following the College cohort with annual questionnaires. (That practice continued through 1955; it's been every two years since then, with only a few exceptions.) The questionnaires are long, and designed to take advantage of the men's high verbal skills. They inquire about employment, health, habits (vacations, sports, alcohol, smoking), political views, family, and, especially, quality of marriage. 27 They have varied somewhat from decade to decade and from director to director, but inevitably they reflect changes in intellectual climate. Only in 1955, for instance, came the first inquiries about college roommates and girlfriends. A perfect prospective study, of course, would forbid retrospective data like this.

By 1948, the Grant Foundation had made good on its threat to withdraw funding. But Alan Gregg of the Rockefeller Foundation stepped in with the salary for a cultural anthropologist. After discussion with Gregg and his colleagues, Bock and the Study staff decided to rename the project the Harvard Study of Adult Development. (Since 1970, that name has covered both the College cohort of the Grant Study and the Inner City cohort of the Glueck Study.) At this point the

Study was essentially supported by Harvard Health Service funds. But to avoid confusion, letters and questionnaires to the men continued to address them as “Grant Study” members, and the Study has continued to be known that way informally to this day.

Margaret Lantis was the new cultural anthropologist, trained to look for, respect, and study cultural differences. She had a doctorate in her field, and was a vastly more sophisticated interviewer, if a less cozy one, than Lewise Gregory, whose training had taken place around Virginia tea tables. In 1950, Lantis interviewed 205 Study men and their wives in their homes. In addition, in the ten years after the war, 171 men dropped by the Study to visit, and the staff was fortuitously able to glean some current information from them. Clark Heath and Gregory usually did those debriefings.

In collaboration with Charles McArthur, a social psychologist who would soon become director of the Study, Lantis administered the Thematic Apperception Test to many of the men. The TAT is a projective test designed to study affect and relationships; this was a departure from the cognitive focus of Frederic Wells. In those days the TAT was a clinical guide, not a research instrument. Twenty years later (in the 1970s), Charles Ducey, a psychologist who was skilled in projective testing but blind to other Study results, reviewed the men’s TATs and their earlier Rorschachs. While the projective tests provided good portraits and corroborated many of the men’s known personality quirks, they did not help the Study make accurate predictions about the men’s mental health or flourishing at age forty-seven. (That’s an age that keeps coming up because it was the approximate age of the men in their twenty-fifth reunion years; more on this shortly. So was seventy-two, the year they celebrated their fiftieth reunions.) In 1981, however, Dan McAdams, soon to become a distinguished researcher of Intimacy and Generativity (see Chapters 4 and 5), successfully used those same TATs to predict intimate relationships in a manner that an academic journal found worthy of publication. His construct of intimacy motivation could significantly predict success in the two fundamental criteria of the good life so often ascribed to Freud— *Arbeiten und lieben* (to be able to work and to love). 28

In 1953, the President and Fellows of Harvard College declared that unless additional support could be garnered from private sources, Harvard would no longer continue to provide for the Study as it had over the five lean years past. And so on July 12, 1954, Drs. Bock and Heath wrote a sad letter: “To all Grant Study Participants, We regret to inform you that financial support of the Study ceased July 1, 1954. . . . Your cooperation has been a source of great satisfaction and pleasure to us. . . .” 29

Final farewells were averted, however. In December of that year, the desperate Study submitted a formal grant to the Tobacco Industry Research Committee for \$15,880 (roughly \$150,000 in 2009 dollars), suggesting that something might be learned about the “positive reasons” that people smoke. And for more than a decade tobacco became the Study’s main source of support, although the documentation is sketchy.

It’s worth noting that in 1954, when this *deus ex machina* materialized, the Study was already sixteen years old. It had achieved the fifteen years first envisioned for it and more, yet there was never any thought of letting it go without a struggle. Longitudinal studies were becoming better known and better understood; there was a lot of motivation to keep it going as the stakes of lifetime studies began to come into sharp focus.

Also in 1954, Dana Farnsworth, M.D., a psychiatrist committed to the mental health of students, replaced the retiring Arlie Bock as director of the Department of Hygiene, now renamed Harvard University Health Services. Frederic Wells had retired. Margaret Lantis had left in 1952 without replacement. In 1955, Clark Heath left the Study due to its financial uncertainty, and Farnsworth appointed Charles McArthur director of the Grant Study.

THE CHANGING OF THE GUARD

With the appointment of a social psychologist as director, the first great phase of the Grant Study came to an end. The early researchers had theorized that development was driven by intellectual ability, physical constitution, and personality traits that invariably resulted from them. This model was now abandoned in favor of a model of relationship-driven development. Years later, when I first became director, Arlie Bock, who had not even included psychiatrists on the full-time staff of his Department of Hygiene, was surprised that I had discovered among his “normal” Grant Study boys alcoholics, personality disorders, and manic depressives. With tongue in cheek he complained to me in the hallway, “They must have been spoiled by you psychiatrists. They didn’t have problems like that when I was running the Study!”

Dana Farnsworth had attended a one-room school in West Virginia, and graduated from Harvard Medical School in 1933. He had headed the health services at Williams College and M.I.T. He was widely considered the nation's foremost expert on student mental health, and was far more sympathetic to psychiatry than Bock had ever been. As soon as he took the helm he hired five full-time and two part-time psychiatrists, and established relationships between them and each of Harvard's residential houses and graduate schools. While Farnsworth gave little direct financial help to the Study, he did provide free office space, and was personally supportive of Charles McArthur and then of me.

McArthur, a brilliant young Health Services psychologist, was completing his Ph.D. in social relations at Harvard when Farnsworth appointed him director of the virtually penniless Study. He held that position from 1955 until I, a research psychiatrist, succeeded him in 1972. When he began, he was the only member of the Study staff, and Harvard was paying him to be a clinician to the students, not to do psychological research. Between 1955 and 1967 he did manage to achieve continuity of funding by asking questions about smoking habits—"If you never smoked, why didn't you?"—a nod to another tobacco-related patron, Philip Morris. And even though he sometimes had to supplement such outside funds with his own staff psychologist's salary, he managed to carry on the work of maintaining addresses, keeping contact with the men, and sending out questionnaires. Between 1960 and 1970, however, there were only four publications, frequency of questionnaires dropped to every three years, and seventeen Grant Study men were noted to have been AWOL for more than a decade.

McArthur's research foci were smoking habits and social class differences among the Grant Study men. He was also interested in longterm validation of the Strong Vocational Test—at the time the best predictor of career choice and satisfaction available—and published several papers on his findings (see Appendix F). This was the first time that the Grant Study really capitalized on its prospective longitudinal design. 30

In 1967 the Grant Foundation awarded McArthur a three-year grant of \$100,000 (\$555,000 in 2009 dollars) to bring Lewis Gregory Davies, now married, back to the Study, and to obtain secretarial help in mailing out questionnaires. Davies's return worked wonders. The seventeen nonresponders came back to the fold. In the forty-five years since then, only seven active members have withdrawn from the Study. No member has been entirely lost to follow-up, for publicly available records and alumni reports have allowed us to characterize the occupational success, marital status, and general life adjustment of those seven dropouts and the twelve who left before 1950. Only four of the nineteen dropouts are still alive; the Study has death certificates for thirteen of the other fifteen.

LATER DATA ANALYSIS CONTINUES: THE TWENTY-FIFTH REUNION YEAR

I was in my first years with the Study when the twenty-fifth reunion years of the College cohort began to roll around. These were the occasion for a major round of data analysis in which we considered the men's adaptation to post-college life and above all to life after the Second World War. The reunions were also an opportunity to compare the College men with their classmates. In 1969 I devised a questionnaire that the Reunion Committee of the Harvard Class of 1944 distributed at the time of its twenty-fifth reunion.

We already knew from freshman physicals that the Grant Study men did not differ significantly from their classmates in such attributes as height, visual acuity, eye color, hay fever, or history of rheumatic fever. Twenty-five years later, it appeared that they did not differ significantly in future occupation, either. A quarter of each class became lawyers or doctors; 15 percent became teachers, mostly at the college level; 20 percent went into business. The remaining 40 percent were distributed through such other fields as architecture, accounting, advertising, banking, insurance, government, and engineering. There were a few artists of varying persuasions. The same proportions held true of the Study men.

But by the reunion years, critical differences had appeared, and some of them are summarized in Table 3.3 . The comparison isn't perfect. Alumni who have not enjoyed relatively smooth sailing in life, love, and career often decline to complete reunion questionnaires. While 92 percent of the Grant Study subjects completed this one, in the table they are being compared to a 70 percent sample of their classmates, probably self-selected for health and success. Even with this selection bias, however, the Study men seemed to have striven harder for success in college and in life than their classmates. Their mean IQ of 135 was only minimally higher than the 130 of their classmates, but a far higher percentage of them graduated with honors. They tended to take fewer sick days. A point of interest: this 1969 analysis was the first time a computer was used to analyze Grant Study data. At that time, the Harvard computer (quite a concept now!) filled a whole building. Data had to be submitted on eighty-column punch cards, often carried half a mile through the snow to the

computing center, and data turnaround could take several days.

Table 3.3 Responses to a Questionnaire Distributed at the 25th Reunion of the Harvard Class of 1944

Very significant = $p < .001$; Significant = $p < .01$; NS = Not significant.

Given their apparent ambition and greater striving, it wasn't surprising that the Grant Study men were more likely than their classmates to have exceeded the success of their fathers by the time they were in their late forties. However, the selection of the Grant Study men favored conventional definitions of success. Stoics outnumbered Dionysians in the Study; the achievement bias in the selection process probably worked against stable youngsters who were more interested in private contentments than in chasing brass rings, and artists whose development takes longer and is frequently less remunerative. Furthermore, capacity for intimacy had been valued less highly in the selection process than the capacity to grin and bear it. One staff member defined a "healthy" person as "someone who would never create problems for himself or anyone else," and many of the Grant men lived up to that ideal. One boasted that what he enjoyed most in life was "being beholden to no one and helping others."

Even taking this into account, the extent of the men's successes as they entered middle age was striking. Four members of the College cohort ran for the U.S. Senate. One served in a presidential cabinet; one was a governor, and another was president. There was a best-selling novelist (no, it was not Norman Mailer, even though he was Harvard class of '43), an assistant secretary of state, and a Fortune 500 CEO. And while the average man in the College cohort had the income and social standing of a successful businessman or physician, he displayed the political outlook, intellectual tastes, and lifestyle of a college professor. At age forty-five, the Grant Study men's average income was about \$180,000 (in 2009 dollars) a year, but fewer than 5 percent of them drove sports cars or expensive sedans. Despite their economic success, they voted for Democrats more often than for Republicans, and 71 percent viewed themselves as "liberal."

I JOIN THE GRANT STUDY AND RE-INTERVIEWS BEGIN

I had arrived at the Grant Study in 1966, supported by an NIMH Research Scientist Development Award. McArthur allowed me to design the next two regular questionnaires. Perhaps the most important change I instituted at the time was to begin collecting physical exam data every five years, including chest x-rays, EKGs, and standard blood and lab work. The exams began as the men turned forty-five (1965–1967) and have continued since then, through age ninety (2010–2012). This prospective record of objective physical health distinguishes the Grant Study from the other great longitudinal studies of personality development.

My approach differed from Charles McArthur's. I wasn't a social psychologist, but an M.D., and in psychoanalytic training. My early studies of recovery from heroin addiction and acute psychosis had left me impressed by the involuntary adaptive coping mechanisms that resilient men employed. My youthful preference for blacks and whites and either/ors had given way to an appreciation of—and a serious interest in—the reality and power of incremental adaptation. Now, at the Grant Study, I could investigate these mysteries to my heart's content.

The questionnaires, concrete as they were, were an invaluable source of information. The men's idiosyncratic responses to standard questions revealed the adaptive styles and behaviors that colored all the other facets of their lives. As one man put it, "We reveal ourselves whenever we say anything." One man sent in a questionnaire two years late because he had just found it under his bed; I wasn't surprised to find evidence of passive aggression in other areas of his existence. We didn't have to depend on words or dream reports, because (as I'll show in Chapter 8) the Study's prolonged and repeated opportunities for observation allowed us to see so-called ego defenses made tangible in concrete behavior. During the years that we overlapped, Charles McArthur was an enthusiastic supporter of this interest.

In 1967 I also began obtaining two-hour interviews of the men, because no questionnaire can convey a person's flavor the way a face-to-face meeting can. This was the first systematic interviewing since Margaret Lantis's tenure in the early 1950s, and the new practice of re-interviewing about every fifteen years has continued through my directorship up to the present.

My second major research interest, maturation, came to life shortly after I joined the Grant Study. My father died when he was forty-four and I was ten. I retained an indelible memory from 1947, my thirteenth year, of his twenty-fifth reunion book, where photographs of barely post-adolescent college seniors sit next door to the portraits of mature

forty-six-year-olds. When the Grant Study men returned for their twenty-fifth reunions beginning in 1967, I experienced my interviews with them as eye-openers; I myself was still a callow thirty-three, but I understood right away that I had unconsciously been waiting for this encounter for twenty-five years. After that, my interest in adult maturation competed with my interests in defense and resilience.

The single most personally rewarding facet of my involvement with the Grant Study has been the chance to interview these men over four decades. The magic of transference hasn't hurt. I am fifteen years younger than they, and when I began, I could not forget that I had been in kindergarten when they had entered the Study. I wanted to call them all "Sir." But they invariably treated me as respectfully as college sophomores would treat a Study physician. I met no condescension, and because I already knew so much about them, the interviews were often remarkably intimate. But they reinforced a belief that the questionnaires had already inculcated in me: that however they may try, people can never neutralize their personalities.

There was an intensity to many of the interviews that was both gratifying and surprising. Often talking with these men was like resuming an old friendship after a period of separation, and that made me feel a little guilty, for I had done little to earn the warmth and trust that they offered. But I soon discovered that whether the men liked me or I liked them had less to do with me than with them. The men who found loving easy made me feel warmly toward them, and left me marveling at my tact and skill as an interviewer. In contrast, the men who had spent their lives fearful of other people and gone unloved in return often left me feeling incompetent and clumsy, like a heartless investigator, vivisecting shy innocents for science.

With some of the men the interviews felt like psychiatric consultations; with some like newspaper profiles; with some like talks with an old friend. I learned to associate a man's capacity to talk frankly of his life with positive mental health. With maturity comes the capacity and the willingness to express emotion in meaningful words.

I learned quickly that the men's responses to me paralleled their ways of relating to people in general. One man, for example, evaded the first few questions I asked him, and then turned to me and said expansively, "Well, let's hear about you!" At first I thought I had just been inept, but then I saw a comment from a staff psychiatrist in 1938: "This boy is more difficult to interview than any I have encountered in the group." One of the warmest, richest personalities in the Study invited me to his home for breakfast at 7 a.m., cooked me a soft-boiled egg, and then extended the interview well past the two hours that I had requested. This in spite of the fact that he was working a sixteen-hour day, that he was moving his entire household to New York in two weeks, that his son was graduating from high school in eight hours, and that he himself had just suffered a devastating business reversal that was front-page news. Far less busy but more socially isolated men would put me off for a week and then meet me in the most neutral setting possible—two of them chose airports!

Some men came to Cambridge to be interviewed, but in most cases I went to them—to Hawaii, Canada, London, New Zealand. One man—only one—seemed very reluctant to be interviewed. But once the interview began he allowed it to extend through his lunch hour, and gave me a lengthy, exciting, and startlingly frank account of his life. Other men readily agreed to see me, but retreated behind ingenious obstacles. Two took every opportunity to interpose their large families between themselves and me when I visited, while others kept their families well out of sight the whole time I was there. There were cultural differences, too. All the New Yorkers and most of the New Englanders saw me in their offices, and few offered me a meal. Virtually all the Midwesterners saw me at home and invited me to dinner. The Californians were evenly divided. Several wives were openly suspicious of the whole enterprise. One spoke so stridently into the telephone that sitting across the desk from her husband I could hear her refuse to see "that shrink" under any circumstances.

Between 1967 and 1970, I re-interviewed a random 50 percent sample of the Study classes of 1942–1944. In 1978–1979, Eva Milofsky, the social-worker daughter of Sigmund Freud's personal physician Max Schur, re-interviewed the rest. Most of the surviving men were interviewed again as they reached retirement age around 1990, almost half by me and the rest by the very gifted Maren Batalden, M.D., whose work will appear in more detail in coming chapters. When the surviving men were about 85 (2004–2006), they, and their wives, were interviewed once more by Robert Waldinger and his team. The Grant Study is probably unique in having obtained so many interviews over so many years. And since the administrative reorganization in 1970, the Study has managed to track the men from the Glueck Inner City

cohort in a fashion virtually identical to our follow-up of the College cohort; both cohorts are now included as subjects in any major Study publication.

To keep reliably in touch with two cohorts over many years is a tough business, and for the last twenty years that business has been in the hands of Robin Western, MFA, our reincarnation of Lewise Gregory. Western has been a tactful and insightful interviewer and a skilled detective, finding our strays and luring them back to the Study. She has been a meticulous and skilled archivist, maintaining seventy years of collected data in manageable order. She's been a key research colleague, planning every questionnaire and wheedling the men's physical exams (with the men's written permission, of course) out of their doctors' offices and into ours. To top it all off, she's been a wonderful friend. If the Grant and Glueck Studies are now among the longest in the world, much credit goes to the perseverance and thoroughness of Robin Western.

Every time I settle down to analyze this wealth of material, yet another of the many richnesses of longitudinal studies becomes clear. No single interview, no single questionnaire, is ever adequate to reveal the complete man, but the mosaic of interviews produced by many observers over many years can be most revealing. One member of the Study, for example, was always seen as dynamic and charismatic by the female staff members, but as a neurotic fool by the males. One shy man from a very privileged background came across to a staff member from a similar milieu as charming, but a colleague from a working-class family thought he was a lifeless stick.

Some concealed truths emerged only after the Study members had been followed for a long time. One reticent man was thirty before he revealed that his mother had had a postpartum depression following his birth. This had not emerged either in his psychiatric interviews at nineteen or in Lewise Gregory's family interview. One man kept his homosexuality secret until he was seventy-five, another until he was ninety. In general, few of the men acknowledged a wife's alcoholism until they were over sixty-five. They were far more honest about their own alcoholism, extramarital affairs, and tax evasion.

In 2005, Robert Waldinger, M.D., a research psychiatrist with a particular interest in intimate relationships, succeeded me as director of the Study. From 2004 to 2006 he re-interviewed and, even better, videotaped consenting married couples and obtained their DNA. In recent years, some of the men have consented to fMRI studies, in an attempt to clarify the positive emotions involved in intimacy. I'll illustrate some of his work in Chapter 6. Some men have donated their brains to the Study. These are further generous sowings whose value will likely not be appreciated for many years.

SOME NOTES ON FUNDING

Keeping the Grant Study in continuous funds for forty years was an anxiety-provoking challenge, but one that with flexibility, resourcefulness, and some sheer good luck I was able to meet.

Since the Career Research Scientist Award that provided my salary for thirty years included no research funds, in 1971 I wrote to the Grant Foundation requesting \$800 to pay the retired Lewise Gregory Davies, now in her seventies, to reenlist recalcitrant Study members once more. At that point we had perhaps seventeen nonresponders. Douglas Bond, former staff psychiatrist to the Grant Study (1942), chairman of the Grant Foundation, dean of Western Reserve Medical School, and son of the Earl Bond who had been advisor to both Arlie Bock and William Grant, came to visit me at Harvard. He listened to my ardent plea on behalf of this uniquely valuable longitudinal study, and then he wrote us a check for \$50,000 (\$280,000 in 2009 dollars). This astonishing munificence provided the seed money for the grant requests and infrastructure by which we obtained continuous funding for the Grant Study for the next three decades. The timing was fortuitous, because the National Institute on Alcohol Abuse and Alcoholism had just opened, and money was relatively plentiful.

Our funding from the NIAAA between 1972 and 1982 required us to pay explicit attention to alcoholism. It also supported the questions in which I had been primarily interested—involuntary coping, relationships, and adult maturation—but the study of alcoholism proved so fascinating that I have given it a chapter of its own (Chapter 9).

The first publication to emerge during my directorship, when the men were about fifty, focused explicitly on prescription drug abuse by the forty-six Study men who became physicians. I used responses from questionnaires and interviews to contrast the frequency of mood-altering drug use and abuse by these men with use and abuse by other Study participants. In this case, the Study's demographic homogeneity presented a convenient means of obtaining a matched

control group for the physicians. The physicians were twice as likely to use and abuse mood-altering drugs as the controls. 31

The publication of this paper in the *New England Journal of Medicine* brought attention to the Grant Study, and it also helped me address an important issue in longitudinal research: whether the process of being studied alters the course of participants' lives. Would the Grant Study doctors, most of whom read the *Journal*, change their self-prescribing behavior in response to my paper? Sadly for them, but happily for my trust in the validity of prospective design, we found ten years later that mood-altering drug abuse had actually increased among the physicians, but not among the controls (who were much less likely to have read the article). Despite my prediction that the doctors would modify their habits (or at least their answers), their misuse of drugs actually got worse. The controls' did not. This result also confirmed my impression that the men were being remarkably honest in their questionnaires.

In 1983, I published *The Natural History of Alcoholism*, which summarized years of research. 32 That same year, to put our next research phase on a more secure financial footing, I accepted an endowed chair in psychiatry at Dartmouth. This meant that Dartmouth would pay my salary, and scarce research funds could be reserved for purposes more interesting than keeping me housed and fed. For ten years the Study moved with its files to Hanover, although it remained the administrative responsibility of Harvard University Health Services.

In 1986 the focus of the Study turned to aging. I was still primarily interested in adaptation, but I was a true Willie Sutton of a researcher, going where the money was. As with the alcohol studies, though, once I learned a little about the subject, I found myself fascinated. And, of course, the matter of aging is absolutely intrinsic to a study of this kind—and to my interest in adult maturation—which was very soon brought vividly home to me.

My first application to the National Institute on Aging was turned down cold. The reason was simple. The proposal's age-phobic fifty-one-year-old author had offered to follow the men's aging prospectively—as progressive decline. (I'll say in my own defense that Shakespeare thought the same way in his youth.) The vigorous seventy-eight-year-old chairman of the NIA research grant committee, eminent gerontologist James Birren, made it abundantly clear that that was not how he viewed aging, and he denied my grant request in no uncertain terms. Forgiving and generative, however, he also undertook to raise my consciousness; and thoroughly chastened but now able to envision aging as a positive process, I rewrote the application. The NIA agreed to fund us, and in 2002 the resulting book, *Aging Well*, captured the inspirational reality of aging as James Birren both imagined and exemplified it. It was a study of the aging process in real time, and it put to scientific rest the negativity of mid-life popular "experts" on aging like Simone de Beauvoir and Betty Friedan. The National Institute on Aging continues to support the Study to this day, and I hope that Chapter 7 will persuade any readers who happen to doubt it that there are worse things in life than living to ninety.

In 1992, Brigham and Women's Hospital offered me secure funding, and the Grant Study came home to Boston. It remained at Brigham and Women's until 2010, when Robert Waldinger, director since 2005, moved it to the Massachusetts General Hospital.

PERSPECTIVES—THEIRS AND OURS

We at the Study have often been asked what effect the men's membership in it has had on them. When we've asked them, most have said that it had no direct effect on them, and indeed there is little to suggest that it has changed their lives in any major way. But as a group they have enjoyed their participation. In 1943 one man wrote, "The Study impressed me by its thoroughness, its interest in the little things, its ability to make the subject feel a part of it rather than a guinea pig. Now it is a gratifying reassurer and friend." A second very shy Study member, who had asked the comely but six years older Lewise Gregory to go to the theater with him (she went), wrote, "I feel my friendship with you (the Study) has more than paid back anything that I put into it." Yet another man wrote, "The very act of being a participant, receiving and pondering the follow-up questionnaires, etc. . . . has made me more self-consciously analytical about my personal development, life choices, career progress, and the like."

Staff members, of course, have our own answers about how the Study has affected us, and I will illustrate here one effect that the Study had on me. Here follows the story of my engagement with one Study member, Art Miller. It's a story of detective work, of which there is plenty in an enterprise like this. It's a story of adaptation. But more than anything, for me it was an object lesson about the dangers of judgment. Precipitous conclusions are a constant danger to incautious

scientists, and the story of Art Miller is my best reminder that a long perspective is our only true protection against it.

THE STORY OF ART MILLER

In 1960, Art Miller disappeared. He came back safely from the war, and told John Monks that he had seen no heavy combat. He went to graduate school, earned a Ph.D. in Renaissance drama, and became an English professor. His published scholarly essays can still be downloaded with Google's help. But then he vanished. For twenty years he was the only "lost" member of the Study. It wasn't until 1980 that I managed to reach his elderly mother, who told me that he had quit his university job long ago and moved to Western Australia, where he was raising his family and teaching high-school drama. I called telephone information in his small town. They sent a man out to his house on a bicycle (things are different in the Outback), only to find that he had moved. The operator called up the headmaster of Miller's school in the middle of the night so as to be able to call me with the new phone number when it was daytime in Boston. When I called Miller at home and told him that I might be coming to Australia, he sounded delighted and insisted on giving me supper and a bed for the night. This evasive man seemed genuinely interested in seeing me.

When I got to Melbourne and called to schedule a meeting, it was the end of the school term, a difficult time for Art. Nevertheless, he agreed to a visit. When I arrived, his front door was open. A note revealed that he was still at the school play, but a nice dinner was laid out and a cold beer was waiting for me in the icebox. Miller and his wife got back after I had been there about an hour, and as the interview proceeded, we made a significant dent in the bottle of Scotch that I had brought as a house present. He had told me that he had moved to Australia out of disaffection with the United States. For one thing, he was afraid that drugs would endanger his children's adolescence; he had not liked "the cut" of their friends. For another, he was committedly against the war in Vietnam. There were some nonspecific reasons for his alienation as well; as he put it, "There was so much anger in the air." I found myself wondering briefly if perhaps he had been running from something.

On the phone calls that followed my visit, Miller was always courteous to me. He never returned another Study questionnaire, however, and he never revealed his whereabouts to Harvard. The last time I reached his number, his daughter told me that he had died of cancer.

I puzzled over Miller. Psychiatrists were now seeing PTSD in many of the Vietnam vets, and Miller had been in a war too. But I quickly dismissed this notion. He had reported "combat fatigue" once during the war; had he perhaps exaggerated a bit? The Woods traits attributed to him in college had been Cultural, Ideational, and Creative and intuitive, and after all, he was a drama professor. In his interview with Monks, he had said that he had not seen sustained combat, and by his own report many of the Grant men had more severe and more prolonged combat histories than he.

The years passed. I formulated various hypotheses, not all of them flattering, for Miller's elusive behavior. In 2009, I had to reread his chart for this book. There I discovered a scrap of paper entered by Clark Heath, who had traveled to Washington in 1950 to review the men's military medical records. Fifty-five years after World War II ended, I learned what had happened to Art Miller in the army. Here is his record from an Italian field hospital, beginning on June 13, 1944:

Patient saw 3 or 4 days of combat, remembers killing three Germans. The last he remembers is attacking up hill with men falling, nearby land blasts, then woke up here two days ago. He has no idea as to what happened in the intervening time. On admission, he was acutely disturbed, kept his fists clenched, and threw himself about, calling "Shells! Bombs! I'm afraid!" and no contact could be established.

There was no change on transfer here. He was restless, disturbed, over-responds to minor stimuli, crawls under covers and into fetal position at sounds of planes and there is no response obtained on questioning.

After two electro-convulsions, amnesia and agitation entirely cleared. There is some noise sensitivity, battle dreams, and fear of combat remains. He is a mildly shy, sensitive, personality type with a strong sense of duty. . . . From an emotional standpoint he will be of no use whatsoever under combat. . . . I feel that he should be reclassified to limited service, permanent.

July 3, 1945: Reclassified. Normal now.

July 5, 1945: Certificate of honorable discharge. Character excellent.

It had been easy to stand in judgment. I could point to Miller's noncompliant and passive-aggressive behavior. I could note that he had run away from his family and his country, and that his earned income was as low as any man's in the Study.

Or I could—and in 2010 I finally did—understand Art Miller's whole life as a creative example of posttraumatic growth. Many playwrights (Edward Albee and Eugene O'Neill among them) have endured traumatic childhoods or major depressions, and spun gold from this wretched straw. It's hard to imagine that Art Miller's students scorned him as a dropout or a victim; on the contrary, they must have greatly enjoyed the drama that this published scholar brought to their small-town school. But it took forty-five years for me to see that truth. And I saw it not by looking back on what Miller had or hadn't done, but by discovering the prospective entry that one very caring physician had made years before, when the trauma was still fresh and contemporary.

A SELF-PORTRAIT

Since it is I who guided the Study for more than half of its existence, it is fitting that I share some relevant aspects of my own development and the potential biases that grow out of them. I was born in New York City to academic WASP parents and educated at Phillips Exeter Academy and Harvard College, where I majored in history and literature and was an editor of the *Lampoon*. I went on to Harvard Medical School with the internalized injunction that teaching and service were good and that business and private practice were bad. And perhaps I put a more than necessarily puritanical spin on this dichotomy, because I remember mocking my professors behind their backs for preferring research to clinical care. I have always considered the New York Times to be the source of truth where news is concerned, and politically I usually vote Democratic. Between 1960 and 2009 I lived mostly in Cambridge, Massachusetts. I was happily married during most of those years, despite more than one divorce. As I write this book I am recently remarried once again, living in southern California, and learning (reluctantly) to broaden my political perspectives.

As a psychiatrist, and like many of my colleagues, I pretend to belong to no "school," but it won't have taken long for my readers to realize that I am trained as a psychoanalyst and am a staunch admirer of Adolf Meyer and Erik Erikson. I began psychoanalytic training not because I thought psychoanalysis "cured" people, but because my best clinical teachers were psychoanalysts. All my life I have wanted to be a teacher, and I have always tried to teach my readers how to look for what transpires invisibly beneath their awareness—in their defenses, in their development, and in their hearts. Less obvious, but also important, is the fact that I worked for two years in a Skinnerian laboratory. There I came to believe that the experimental method far surpasses intuition as a means of uncovering truth, and that by charting behavior over time, patterns may be revealed that would interest the staunchest of psychoanalysts. Despite the strictures of my medical student conscience, I have spent more time in "selfish" research pursuits than in "noble" clinical care.

I should probably note that my interest in Alcoholics Anonymous (see Chapter 9) does not have to do with being in recovery myself. Here again I was following Willie Sutton's example. To run the Grant and Glueck Studies I needed a Harvard appointment. The only job opening at the time was a co-directorship of the Alcohol Treatment Center at Cambridge Hospital. As a condition of employment at the Treatment Center I went to AA meetings once a month for ten years. I came to scoff but stayed to admire, and as a result of my enthusiasm in 1999 I was appointed a Class A (that is, non-alcoholic) trustee of AA, and the six years that I spent in that capacity were among the best in my life.

I have always loved big questions. At ten I wrote my sixth-grade term paper on the origin of the universe. I've already told how mesmerized I was, two years after my father's death, when our family received a courtesy copy of his twenty-fifth Harvard reunion book. I marveled at how those raw college caterpillars had evolved into mature forty-seven-year-old men, and was first struck by the profound implications of adult development.

Until I was eighteen, I intended to become an astrophysicist. By nineteen, however, I had read my roommate's copy of Robert White's *Lives in Progress*, perhaps the first prospectively designed textbook on adult development, and decided to go to medical school instead.³³ In my training and in the years that followed it, I was fascinated by people's capacity to recover from apparent catastrophe and continue to develop over the course of their lives. I became a dedicated believer in the power of long-term prospective study to answer psychiatric riddles, and I read avidly about the world's great longitudinal studies. They entranced me; they were telescopes, all right, but the kind that could focus on human lives, not just inanimate stars. I was equally entranced by B. F. Skinner's idea of the cumulative record, a sort of behavioral cousin of the EKG that draws a "picture" of people's doings as they change over time, allowing us to visualize those changes in

a single gestalt. In 1966, while still an assistant professor of psychiatry at Tufts Medical School, I joined the Grant Study and was immediately hooked.

My dream was to make the Grant Study, at the time little known, as important as the longitudinal studies that had so inspired me. One of the few memories I have of my training psychoanalysis was holding up my brand-new key to the Study files and crowing to my analyst, “I have acquired the key to Fort Knox!” But even I did not anticipate that I would still be working on the Study full-time forty-five years later.

With financial support from an NIMH Research Scientist Development Award, I worked on the Grant Study for five years before moving my appointment as Associate Professor of Psychiatry from Tufts to Harvard, and succeeding Charles McArthur as director of the Study.

At the time that I took over, Harvard was still not convinced of either the power or the importance of the Study, and was seriously considering destroying the sensitive files that so enthralled me. (They were especially sensitive just then because the year before, protesting students had broken the windows of the Health Services and occupied Harvard’s administrative offices.)

In 1973, I gave a dinner as the new director. I invited various Harvard potentates and Radcliffe president Matina Horner, the sole woman. I described the Study’s plight, and while I was still resisting entreaties by the Harvard authorities to reduce the case records to soulless microfiche, Horner established at Radcliffe the Henry A. Murray Center for the Study of Lives—a center dedicated to preserving longitudinal studies in their original form. One of its first and most exciting acquisitions was the records of the Harvard Study of Adult Development. Over the years (1976–2003) of the Murray Center’s existence, its director, Ann Colby, was an inspiration to all who knew her, especially me.

The focus of the Study shifted again as my interests pushed it away from sociology and toward epidemiology and psychodynamics. I was especially fascinated by involuntary coping mechanisms and by the relationship between psychological stress and physical symptoms. I am increasingly convinced, especially since passing seventy myself, that psychiatry and psychology need to become more aware of people’s positive emotions and experiences of spirituality. I believe too that a predilection for love and compassion are hard-wired in mammals. One early reviewer of this book said to me, “George, your view of adult development is sooooo 1970s.” But you know what? The seventies weren’t so bad. Those were the years of Bernice Neugarten, Robert Kegan, Jane Loevinger, Emmy Werner, and Jerome Kagan, and I do not believe that their work has so far been surpassed. And anyway, like all the Study directors, like all the men whose life voyages we accompanied, and like the Grant Study itself, I was—I am—a creature of my time. And from age thirty-three to age seventy-eight my participation in the Study has been an abiding joy.

INTO THE FUTURE

One of the points I’ll be making throughout this book is that our close relationships when we are young make an enormous difference in the quality of our lives. Yet there have been few studies of intimate relationships near the end of life, let alone of how these relationships affect physical and mental health. In 2003, Robert Waldinger began to study the marriages of the Grant Study men, now in their tenth decades, and that meant inviting their wives to join the Study. Waldinger, associate professor at Harvard Medical School and a psychoanalyst, has since succeeded me as director. His fresh view of the project is now giving the Study a microscopic (as opposed to my telescopic) look at the daily lives of single and married older adults.

However, the National Institute of Mental Health has not been willing to fund such studies into the twenty-first century. Scientific fashion has once again abandoned the social for the biological. If Willie Sutton wanted grant money to investigate adult lives today, he’d have to do brain studies. Since 2005, Waldinger has obtained funding for neuroimaging studies from three sources: the Harvard Neurodiscovery Center, the Fidelity Foundation, and once again the National Institute on Aging. And for the third time since it stopped funding the Study in 1945, the W. T. Grant Foundation has come to the rescue with additional funds.

These vicissitudes of funding may be nerve-racking, but they are not an unmitigated curse. Over the last five years, they have encouraged the Study to add biological and neuroscientific data to nearly seventy-five years of information about men’s lives. Most surviving Grant Study men are allowing the Study to collect their DNA, conduct sensitive tests of their intellectual capacity, and carry out neuroimaging of their brain structure and function to create an unprecedented

and irreplaceable resource that someday may help us understand the links between brain and behavior in human aging. The combination of life histories and neuroscience data will allow us to investigate fundamental and pressing questions about the aging process. For example, how do genes and environmental factors—such as alcohol use or exposure to traumatic stress—interact to determine who maintains an active and vigorous mind and body into their nineties and who does not? Are there lifestyle choices and behaviors that can buffer the human brain against the ravages of old age?

As the population grows older, these questions become more urgent for families, for healthcare providers, and for public policymakers. As I've noted before, there are advantages as well as disadvantages to studying an elite sample. One of the advantages is that the unusually healthy Grant Study ninety-year-olds offer some insight into what life will look like in thirty years for the average baby boomer. The seventy-five-year-old Study may look dated to modern (or very young) investigators. But it is well to recall Santayana's warning: "Those who cannot remember the past are condemned to repeat it."